

Please check that this question paper contains 09 questions and 02 printed pages within first ten minutes.

EVENING

[Total No. of Questions: 09]

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Uni. Roll No.

16 MAY 2024

Program: B.Tech. (IT)

Semester: 3rd

Name of Subject: Digital Circuits and Logic Design

Subject Code: ESIT-101

Paper ID: 16042

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Solve the following $(87.125)_{10} = ()_{16}$
- b) State basic theorems of Boolean algebra.
- c) Define Prime implicants & essential Prime -Implicants
- d) How are sequential statements different from concurrent statements in VHDL?
- e) Distinguish between synchronous and asynchronous counters.
- f) Give 2 examples each of D/A and A/D converter.

Part – B

[Marks: 04 each]

- Q2.** Design 16:1 multiplexer using 8:1 multiplexer.
- Q3.** Simplify Boolean expression $Y = ABC' + ABC + A'B'C$ on K-Map
- Q4.** Simplify function using K-map $F(A,B,C,D) = M(1,3,4,5,6,11,13,14,15)$
- Q5.** Explain characteristic equation of JK Flip-flop from excitation table.
- Q6.** Design a 4 bit ripple counter using T Flip Flop. Explain using wave forms.
- Q7.** Design a 4 bit magnitude comparator to compare two 4 bit numbers

Part – C

[Marks: 12 each]

- Q8.** a) What is Full Adder? Implement Full adder using two half Adders. b) Distinguish between level triggering and edge triggering.

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OR

a) Explain the working of SIPO & PISO register? b) What is a race around condition? How can we eliminate it?

Q9.

a) Why we need A/D techniques? A 4 bit D/A converter have an output range of '0 to 2.5V. Define its resolution b) Draw the block diagram of a 32:1 multiplexer tree and describe how its input is directed to the output in this system.

OR

a) Elaborate the working of Dual Slope type converter. b) Write down the truth table, entity declaration and behavioural/dataflow architecture for the Encoder in VHDL. Also draw the circuit and output waveforms.
